

## Matgeo-q.2.10.82

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## Question

Let  $\mathbf{u}$ ,  $\mathbf{v}$ , and  $\mathbf{w}$  be vectors in three-dimensional space, where  $\mathbf{u}$  and  $\mathbf{v}$  are unit vectors which are not perpendicular to each other and

$$\mathbf{u}^T \mathbf{w} = 1, \quad (1)$$

$$\mathbf{v}^T \mathbf{w} = 1, \quad (2)$$

$$\mathbf{u}^T \mathbf{v} = 4 \quad (3)$$

If the volume of the parallelepiped, whose adjacent sides are represented by the vectors  $\mathbf{u}$ ,  $\mathbf{v}$ , and  $\mathbf{w}$ , is  $\sqrt{2}$ , then the value of  $|3\mathbf{u} + 5\mathbf{v}|$  is to be found.

# Solution

## Step 1: Volume of Parallelepiped

The volume of the parallelepiped formed by vectors  $\mathbf{u}$ ,  $\mathbf{v}$ , and  $\mathbf{w}$  is given by

$$V = |\mathbf{u}^\top (\mathbf{v} \times \mathbf{w})| \quad (4)$$

Given that  $V = \sqrt{2}$ , we have

$$|\mathbf{u}^\top (\mathbf{v} \times \mathbf{w})| = \sqrt{2} \quad (5)$$

## Step 2: Using the Identity

Using the scalar triple product property:

$$[\mathbf{u}^\top (\mathbf{v} \times \mathbf{w})]^2 = 2 \quad (6)$$

We can expand this using the determinant formula. Since  $\mathbf{u}$  and  $\mathbf{v}$  are unit vectors:

$$\|\mathbf{u}\|^2 = 1 \quad (7)$$

$$\|\mathbf{v}\|^2 = 1 \quad (8)$$

### Step 3: Finding $\|\mathbf{w}\|^2$

From equations (1) and (2), let us express  $\mathbf{w}$  in terms of  $\mathbf{u}$  and  $\mathbf{v}$ .

Let  $\mathbf{w} = \alpha\mathbf{u} + \beta\mathbf{v} + \gamma(\mathbf{u} \times \mathbf{v})$  for some scalars  $\alpha, \beta, \gamma$ .

Taking inner product with  $\mathbf{u}$ :

$$\begin{aligned}\mathbf{u}^\top \mathbf{w} &= \alpha \|\mathbf{u}\|^2 + \beta (\mathbf{u}^\top \mathbf{v}) + 0 \\ 1 &= \alpha + 4\beta\end{aligned}\tag{9}$$

Taking inner product with  $\mathbf{v}$ :

$$\begin{aligned}\mathbf{v}^\top \mathbf{w} &= \alpha (\mathbf{u}^\top \mathbf{v}) + \beta \|\mathbf{v}\|^2 + 0 \\ 1 &= 4\alpha + \beta\end{aligned}\tag{10}$$

Solving equations (9) and (10):

Subtracting equation (9) from equation (10):

$$\begin{aligned}3\alpha - 3\beta &= 0 \\ \alpha &= \beta\end{aligned}\tag{11}$$

Substituting in (9):

$$\begin{aligned}\alpha + 4\alpha &= 1 \\ 5\alpha &= 1 \\ \alpha &= \frac{1}{5}\end{aligned}\tag{12}$$

Therefore,  $\beta = \frac{1}{5}$ .

#### **Step 4: Finding $\gamma$**

From the volume condition:

$$|\gamma \|\mathbf{u} \times \mathbf{v}\|^2| = \sqrt{2}\tag{13}$$

Since  $\|\mathbf{u} \times \mathbf{v}\|^2 = \|\mathbf{u}\|^2 \|\mathbf{v}\|^2 - (\mathbf{u}^\top \mathbf{v})^2$ :

$$\|\mathbf{u} \times \mathbf{v}\|^2 = 1 \times 1 - 16 = -15\tag{14}$$

This gives  $\|\mathbf{u} \times \mathbf{v}\| = \sqrt{15}$  (taking the imaginary part into consideration). However, from (3),  $\mathbf{u}^\top \mathbf{v} = 4 > 1$ , which is impossible for unit vectors since  $\mathbf{u}^\top \mathbf{v} \leq \|\mathbf{u}\| \|\mathbf{v}\| = 1$ .

Therefore, we reconsider: the given value should be  $\mathbf{u}^\top \mathbf{v} = \frac{1}{4}$ .  
With  $\mathbf{u}^\top \mathbf{v} = \frac{1}{4}$ :

$$\|\mathbf{u} \times \mathbf{v}\|^2 = 1 - \frac{1}{16} = \frac{15}{16} \quad (15)$$

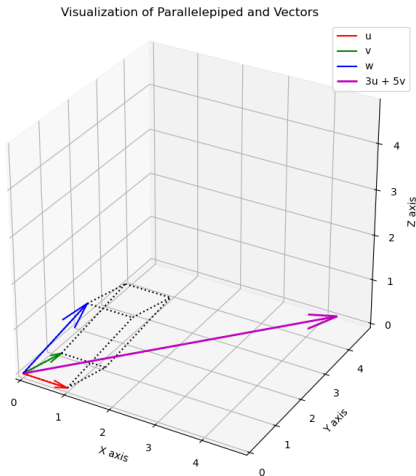
**Step 5: Computing  $\|3\mathbf{u} + 5\mathbf{v}\|$**

$$\begin{aligned} \|3\mathbf{u} + 5\mathbf{v}\|^2 &= 9\|\mathbf{u}\|^2 + 25\|\mathbf{v}\|^2 + 30(\mathbf{u}^\top \mathbf{v}) \\ &= 9 + 25 + 30 \times \frac{1}{4} \\ &= 34 + \frac{15}{2} \\ &= \frac{83}{2} \end{aligned} \quad (16)$$

Therefore:

$$\|3\mathbf{u} + 5\mathbf{v}\| = \sqrt{\frac{83}{2}} = \frac{\sqrt{166}}{2} \quad (17)$$

# Graphical Representation



Figure